MSBA 5223 – Database Foundations
Assignment 2

The purpose of this assignment is to help you practice the following skills that are essential to your success in this course and in your professional life beyond school if you have the opportunity to work with databases.

- Use the Management Studio to run, open, and save SQL queries.
- Create a Database Diagram for an existing database.

Another purpose of this assignment is for you to possibly struggle and feel confused while you apply what you have learned in a slightly different context than what was provided in examples. This will allow you to internalize the use of the various techniques and develop your own approach for addressing problems. The purpose is NOT to make you struggle overmuch and hit your head against a brick wall. If you don't know why something isn't working, ask! If you are not sure why something is doing something other than you expected, ask. I am here to help you learn.

Create a Database Diagram for the CentralCollege Database

Restore the database CentralCollege to your SQL server. You will need to change the ownership to yourself.

1. Create a Database Diagram for CentralCollege that includes ALL tables. Arrange the tables so that all show completely and fit onto 1 page of 8.5 by 11 paper. Ensure that no relationship lines are crossed. **Add a text box in the upper right corner with your name and the name of the database**. **Print this to a PDF.**
2. Submit the document **as a PDF** to Assignment 02.

MSBA 5223 – Database Foundations
Assignment 3

The purpose of this assignment is to help you practice the following skills that are essential to your success in this course and in your professional life beyond school if you have the opportunity to work with databases.

- Use the Management Studio to run, open, and save SQL queries.
- Code SELECT statements that require any of the language elements presented in this chapter or previous chapters.

Another purpose of this assignment is for you to possibly struggle and feel confused while you apply what you have learned in a slightly different context than what was provided in examples. This will allow you to internalize the use of the various techniques and develop your own approach for addressing problems. The purpose is NOT to make you struggle overmuch and hit your head against a brick wall. If you don't know why something isn't working, ask! If you are not sure why something is doing something other than you expected, ask. I am here to help you learn.

Notes about submitting your assignment:

You will create a working document in word processing. **The assignment name, i.e. Assignment 3, and your name must be at the top of every page**. If you do not know how to do this in a header, ask me.

You will see a text box similar to the following each time you should put something in the document.

> **Answer #X**
> **Paste your entire query into Word. Be sure to keep the color-coding.**
> **Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document.**

First, make sure that your answer is numbered in the word document and make sure that all your answers are in order. There are multiple ways to easily do this. I want to see the number of the answer **BEFORE** I see your query and/or screenshot.

When you are asked for the query, copy it from SSMS and paste it into your document. It should keep the color-coding. In MS Word, this will be an option when you are pasting.

When asked for the results, I want to see something similar to the following:

Results | Messages

	InvoiceID	VendorID	InvoiceNumber	InvoiceDate	InvoiceTotal	PaymentTotal	CreditTotal	TermsID	InvoiceDueDate	PaymentDate
1	1	122	989319-457	2019-10-08	3813.33	3813.33	0.00	3	2019-11-08	2019-11-07
2	2	123	263253241	2019-10-10	40.20	40.20	0.00	3	2019-11-10	2019-11-14
3	3	123	963253234	2019-10-13	138.75	138.75	0.00	3	2019-11-13	2019-11-09
4	4	123	2-000-2993	2019-10-16	144.70	144.70	0.00	3	2019-11-16	2019-11-12
5	5	123	963253251	2019-10-16	15.50	15.50	0.00	3	2019-11-16	2019-11-11
6	6	123	963253261	2019-10-16	42.75	42.75	0.00	3	2019-11-16	2019-11-21
7	7	123	963253237	2019-10-21	172.50	172.50	0.00	3	2019-11-21	2019-11-22
8	8	89	125520-1	2019-10-24	95.00	95.00	0.00	1	2019-11-04	2019-11-01
9	9	121	97/488	2019-10-24	601.95	601.95	0.00	3	2019-11-24	2019-11-21
10	10	123	263253250	2019-10-24	42.67	42.67	0.00	3	2019-11-24	2019-11-22
11	11	123	963253262	2019-10-25	42.50	42.50	0.00	3	2019-11-25	2019-11-20
12	12	96	I77271-O01	2019-10-26	662.00	662.00	0.00	2	2019-11-16	2019-11-13
13	13	95	111-92R-10096	2019-10-30	16.33	16.33	0.00	2	2019-11-20	2019-11-23
14	14	115	25022117	2019-11-01	6.00	6.00	0.00	4	2019-12-10	2019-12-10
15	15	48	P02-88D77S7	2019-11-03	856.92	856.92	0.00	3	2019-12-02	2019-11-30
16	16	97	21-4748363	2019-11-03	9.95	9.95	0.00	2	2019-11-23	2019-11-22
17	17	123	4-321-2596	2019-11-05	10.00	10.00	0.00	3	2019-12-04	2019-12-05
18	18	123	963253242	2019-11-06	104.00	104.00	0.00	3	2019-12-05	2019-12-05

Query executed successfully. | TLWILLIAMS-DELL (15.0 RTM) | TLWILLIAMS-DELL\drtlw ... | AP | 00:00:00 | 114 rows

This is very easy to do with Snip & Sketch in Windows 10. Shft-WindowsKey_S will enable it. Notice that it is **only** the **results** area of SSMS. **Notice that the entire status bar is displayed at the bottom**. Sometimes it is not possible

to capture all the rows in one results screen. The status bar gives me the total rows along with other information that I require. If the status bar is missing from your screen shot and I cannot read it across, it will not count as an answer. If all the rows in the result are not in one screen, your screenshot should include the FIRST rows.

Be sure to save your document periodically. When you have finished the assignment, save the document as a PDF.

Submit the PDF to the assignment.

Each answer is weighted equally and is worth 10 points.

Execute Queries to Retrieve Information from the CentralCollege Database

1. Write a SELECT statement that returns all of the columns and all of the rows from the Courses table. Then, execute this statement to make sure it works correctly.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #1.

2. Write a SELECT statement that returns three columns from the Courses table: CourseNumber, CourseDescription, and CourseUnits. Then, execute this statement to make sure it works correctly.

   Add an ORDER BY clause to this statement that sorts the result set by CourseNumber in ascending sequence. Then, run this statement again to make sure it works correctly. This is a good way to build and test a statement, one clause at a time

Paste your entire **final** query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #2.

3. Write a SELECT statement that returns one column from the Students table named FullName that joins the LastName and FirstName columns.

   Format this column with the last name, a comma, a space, and the first name like this:

   Doe, John

   Sort the result set by last name in ascending sequence.

   Return only the students whose last name begins with a letter from A to M. Use the operator for pattern matching to accomplish this.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #3.

4. Write a SELECT statement that returns the following column names and data from the Instructors table. The name should be formatted as in query 3 with the column name of **Instructor**. There should not be any rows with a NULL Instructor name.

   LastName

   FirstName

   AnnualSalary

   Return only the rows with an annual salary that's greater than or equal to 60,000.

   Sort the result set in descending sequence by the AnnualSalary column.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #4.

5. Write a SELECT statement that returns the following column names and data from the Instructors table. The name should be formatted as in query 3 with the column name of **Instructor**. There should not be any rows with a NULL Instructor name.

   LastName

   FirstName

   HireDate

   Return only the rows with a hire date that's in 2019.

   Sort the result set in ascending sequence by the HireDate column.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #5.

6. Write a SELECT statement that returns the following column names and data from the Students table. The name should be formatted as in query 3 with the column name of **Student**. There should not be any rows with a NULL Student name.

   FirstName

   LastName

   EnrollmentDate

   CurrentDate

   MonthsAttended — A column that's calculated by getting the difference between the enrollment date and the current date

   To get the value of the months attended, use the DATEDIFF function with the month argument.

   Sort the result set in ascending sequence by the MonthsAttended column.

Paste your entire query into Word. Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #6.

7. Write a SELECT statement that returns the following column names and data from the Instructors table. The name should be formatted as in query 3 with the column name of **Instructor**.

   FirstName

   LastName

   AnnualSalary

   Return only the top 20 percent of instructors based on annual salary.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #7.

8. Write a SELECT statement that returns the following column names and data from the Students table. The name should be formatted as in query 3 with the column name of **Student**.

   LastName

   FirstName

   Return only the rows where the LastName column starts with the letter 'G' and contains an ‘a’. To do that, use the LIKE phrase.

   Sort the result set by last name in ascending sequence.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #8.

9. Write a SELECT statement that returns the following column names and data from the Students table. The name should be formatted as in query 3 with the column name of **Student**.

   LastName

   FirstName

   EnrollmentDate

   GraduationDate

   Return only the rows where the EnrollmentDate column is greater than 12-01-2019 (Dec 1) and the GraduationDate column contains a null value.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #9.

10. Write a SELECT statement without a FROM clause that creates a row with these columns:

Column	**Source**
Price	100 (dollars)
TaxRate	.07 (7 percent)
TaxAmount	The price multiplied by the tax rate
Total	The price plus tax

To calculate the fourth column, add the **expressions** you used for the first and third columns:

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #10.

11. Make sure your name is on your document. Submit the one document **as a PDF** to the Assignment - Assignment 03.

MSBA 5223 – Database Foundations
Assignment 4

The purpose of this assignment is to help you practice the following skills that are essential to your success in this course and in your professional life beyond school if you have the opportunity to work with databases.

- Code SELECT statements that require any of the language elements presented in this chapter or previous chapters.

Another purpose of this assignment is for you to possibly struggle and feel confused while you apply what you have learned in a slightly different context than what was provided in examples. This will allow you to internalize the use of the various techniques and develop your own approach for addressing problems. The purpose is NOT to make you struggle overmuch and hit your head against a brick wall. If you don't know why something isn't working, ask! If you are not sure why something is doing something other than you expected, ask. I am here to help you learn.

Notes about submitting your assignment:

Instructions on submitting are detailed in assignment 2.

Be sure to save your document periodically. When you have finished the assignment, save the document as a PDF.

Each answer is weighted equally and is worth 10 points.

Execute Queries to Retrieve Information from the CentralCollege Database

1. Write a SELECT statement that joins the Courses table to the Departments table and returns these columns: CourseNumber, CourseDescription, DepartmentName.
   Start with Courses and use table aliases.
   Sort the result set by DepartmentName and then by CourseNumber in ascending order.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #1.

2. Write a SELECT statement that joins the Instructors table to the Courses table and returns these columns: LastName, FirstName, CourseNumber, CourseDescription.
   Use table aliases.
   Return all courses for each instructor with a status of "P" (part time).
   Sort the result set by LastName and then by FirstName in ascending order.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #2.

3. Write a SELECT statement that joins the Departments, Courses for the Department, and Instructors for the Courses. This statement should return these columns: DepartmentName, CourseDescription, FirstName, and LastName (of the instructor for the course).
   Use aliases for the tables and return **only** those courses with three units (or credit hours.)
   Sort the result set by DepartmentName and then by CourseDescription in ascending sequence.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #3.

4. Write a SELECT statement that joins the Departments, Courses (for the department), StudentCourses, and Students tables. This statement should return these columns: DepartmentName, CourseDescription, LastName, and FirstName (of the student taking the course).
   Use table aliases.
   Return all courses in the English department. Do NOT write a separate query to find the DepartmentID of the English department.
   Sort the result set by DepartmentName and then by CourseDescription in ascending sequence.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #4.

5. Read entirely before beginning.
   Write a SELECT statement that returns these columns:

InstructorDept	The DepartmentName column from the Departments table for a related instructor
LastName	The LastName column from the Instructors table
FirstName	The FirstName column from the Instructors table
CourseDescription	The CourseDescription column from the Courses table
CourseDept	The DepartmentName column from the Departments table for a related instructor

   Return one row for each **course** that's in a different department than the department of the instructor assigned to teach that course. (Hint: You will need to join the Departments table to both the Instructors table and the Courses table, which will require you to use table aliases to distinguish the two tables.)

   Start with the **Courses** table.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #5.

Execute Queries to Retrieve Information from the AdventureWorks database

6. List all employees with their BusinessEntityID, last name, first name, job title, and date of birth. Sort the list by their last name, then their first name. You will use **HumanResources.Employee** and **Person.Person.** Use table aliases.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #6.

7. List the top 5 sales people based on the YTD sales. Include their name in the format **lastname, firstname**, i.e. Williams, Terry. Include the SalesYTD and their commission YTD. Make sure all columns have names. You will use **Sales.SalesPerson** and **Person.Person**. Use table aliases.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #7.

8. Read entirely before beginning.
   Create an email list that includes Employee name in the format "Lastname, Firstname" and the employee's email address. The employee name column should be called EmployeeName. The email address column should be called Email. Create this list only for the Executive General and Administration **group**. Do NOT do a separate query to find department ids for this group. Look at the data! Sort it by the employee's last name. You will use the following tables in the following order:

   HumanResources.Employee
   Person.Person
   HumanResources.EmployeeDepartmentHistory
   HumanResources.Department
   Person.EmailAddress

   Before you start, look at the HumanResources.EmployeeDepartmentHistory table. How do you know which department someone is **currently** in? (Look at StartDate and EndDate.) Start working by writing and executing a query that will give you a list of employees and their current department. HINT: This should give you the same number of total rows that you get when selecting all the rows from HumanResources.Employee. Filter this so that you only have the Executive General and Administration **group.**

   Then add in the Person.Person, Person.EmailAddress. Be sure to have them in the order listed above.

   Your answer should have 35 rows.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #8.

9. List the names of all employees, their job title, and date of birth. The names should be in a single column in the format of first name and then last name. There should only be one space between the first and last names. This column should be called **EmployeeName**. Include the employee's first address line, their city, state and postalcode concatenated together appropriately and called **CSZ**, their phone number and the type of phone number. You will use the following tables **IN THIS ORDER**.

   HumanResources.Employee
   Person.Person
   Person.BusinessEntityAddress
   Person.Address
   Person.StateProvince
   Person.PersonPhone
   Person.PhoneNumberType

   This is a total of 7 tables in the query. You may need to view the data in all of the tables, paying attention to column names to determine how to join them. **Use table aliases!**

Paste your entire query into Word. Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #9.

10. Create and execute a new query that will list the BusinessEntityID, FirstName, MiddleName, and LastName of all the employees whose last name starts with an Z and whose first name starts with an E or an I or a N and contains exactly 6 characters. This is considered pattern matching. Do NOT use any functions not covered in class or in the textbook chapters 1-4.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #10.

    Make sure your name is on your document. Submit the one document **as a PDF** to the Assignment - Assignment 4.

The purpose of this assignment is to help you practice the following skills that are essential to your success in this course and in your professional life beyond school if you have the opportunity to work with databases.

- Apply knowledge that you have acquired from your readings or in class
- Code SELECT statements that require any of the language elements presented in these chapters or previous chapters.

Another purpose of this assignment is for you to possibly struggle and feel confused while you apply what you have learned in a slightly different context than what was provided in examples. This will allow you to internalize the use of the various techniques and develop your own approach for addressing problems. The purpose is NOT to make you struggle overmuch and hit your head against a brick wall. Work with your study group! If you don't know why something isn't working, ask me! If you are not sure why something is doing something other than you expected, ask me. I am here to help you learn although I do expect you to think about it and try before asking!

Notes about submitting your assignment:

More detailed information was in Assignment 3.

Be sure to save your document periodically. When you have finished the assignment, save the document as a PDF.

Submit the PDF to the assignment.

Each answer is worth 20 points.

Execute Queries to Retrieve Information from the CentralCollege Database

1. Write a SELECT statement that joins the Instructors and Courses tables and returns these columns: LastName, FirstName, and CourseDescription.

   Return at least one row for each instructor, even if that instructor isn't teaching any courses.

   Sort the result set by LastName and then by FirstName.

Paste your entire final query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #1.

2. Use the UNION operator to generate a result set consisting of five columns from the Students table:

Status	A calculated column that contains a value of UNDERGRAD or GRADUATED
FirstName	The FirstName column
LastName	The LastName column
EnrollmentDate	The EnrollmentDate column
GraduationDate	The GraduationDate column

   If the student doesn't have a value in the GraduationDate column, the Status column should contain a value of UNDERGRAD. Otherwise, it should contain a value of GRADUATED.

   Sort the final result set by EnrollmentDate.

Paste your entire final query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #2.

3. Write a SELECT statement that returns these two columns:

DepartmentName	The DepartmentName column from the Departments table
CourseID	The CourseID column from the Courses table

Return all departments with no courses. (Hint: Use an outer join and only return rows where the CourseID column contains a null value.)

Paste your entire final query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #3.

Execute Queries to Retrieve Information from AdventureWorks

4. Write a SELECT statement that will list all persons who do NOT have the PersonType of IN or EM and that do NOT have a BusinessEntityContact.

   Display these columns:
   The BusinessEntityID column from the Person.Person table
   The PersonType column from the Person.Person table
   The Last Name and First Name columns from Person.Person table.

   You will need the Person.BusinessEntityContact table. Pay close ATTENTION to the relationships between Person and BusinessEntityContact. You want the contacts for the person, not for the Business.

   Sort the result set by the person's last name and then first name.

5. In AdventureWorks, a new manager believes that Special Offers have been underutilized. During a Special Offer meeting, you are asked to create a report about Special Offers NOT assigned to products.

   Use the tables Sales.SpecialOffer and Sales.SpecialOfferProduct to determine if a special offer is assigned (or NOT assigned) to a product.

   Once the Special Offers not assigned to products have been identified, report the following information:

   SpecialOfferID

   SpecialOfferDescription

   SpecialOfferEndDate

   All results must reflect Special Offers that have not yet ended, with discounts greater than 0%. To evaluate ending dates, assume today is January 1, 2012.

   Your answer should have 1 row.

Paste your entire final query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #5.

6. Make sure your name is on your document. Submit the one document **as a PDF** to Assignment 5.

MSBA 5223 – Database Foundations
Assignment 6

The purpose of this assignment is to help you practice the following skills that are essential to your success in this course and in your professional life beyond school if you have the opportunity to work with databases.

- Apply knowledge that you have acquired from your readings or in class
- Code SELECT statements that require any of the language elements presented in these chapters or previous chapters.

Another purpose of this assignment is for you to possibly struggle and feel confused while you apply what you have learned in a slightly different context than what was provided in examples. This will allow you to internalize the use of the various techniques and develop your own approach for addressing problems. The purpose is NOT to make you struggle overmuch and hit your head against a brick wall. Work with your study group! If you don't know why something isn't working, ask me! If you are not sure why something is doing something other than you expected, ask me. I am here to help you learn although I do expect you to think about it and try before asking!

Notes about submitting your assignment:

More detailed information was in Assignment 3.

Be sure to save your document periodically. When you have finished the assignment, save the document as a PDF.

Submit the PDF to the assignment.

Each answer is worth 12.5 points.

Execute Queries to Retrieve Information from the CentralCollege Database

1. Write a SELECT statement that returns these columns:

NumberOfInstructors	Use COUNT to get the number in the Instructors table.
AvgAnnualSalary	USE AVG to get the average of the AnnualSalary for Instructors

Only include the rows where the Status column is equal to "F" (Fulltime)

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #1.

2. Write a SELECT statement that returns one row for each department that has instructors. The statement should return these columns:

DepartmentName	From the Departments table
NumberOfInstructors	The count of the instructors in that department
HighestSalary	The AnnualSalary of the highest-paid instructor in that department.

Sort the results so that the department with the most instructors is first.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #2.

3. Write a SELECT statement that returns one row for each instructor that has courses. The statement should return these columns:

InstructorName	Use CONCAT() to put into the format "John Doe"from the Instructors table
NumberOfCourses	A count of the number of courses for each instructor
TotalUnits	A sum of the CourseUnits from the Courses for the instructor

Sort the results in descending sequence by the TotalUnits.

NOTE: You may need to concatenate the InstructorName again in the GROUP BY clause

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #3.

4. Write a SELECT statement that returns one row for each course that has students enrolled. The statement should return these columns:

DepartmentName	From the Departments table
CourseDescription	From the Courses table
NumberOfStudents	A count of the number of students in the course

Sort the results by DepartmentName, then by the enrollment for each course.

Paste your entire final query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #4.

5. Write a SELECT statement that returns one row for each student that has courses. The statement should return these columns:

StudentID	From the Students table
TotalUnits	The sum of the CourseUnits for each student

Sort the results in descending sequence by the total CourseUnits for each student.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #5.

6. Modify the solution to number 5 so it only includes students who haven't graduated and who are taking more than nine units. Do not change anything else.

Paste your entire query into Word. **Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #6**

7. Write a SELECT statement that answers this question: What is the total number of courses taught by **part-time** instructors only? Return these columns:

InstructorName	Use CONCAT() to put into the format "John Doe"from the Instructors table
TotalCourses	The total number of courses taught by each instructor.

Use the ROLLUP operator to include a row that gives the grand total.

Paste your entire query into Word.
Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #10.

Execute Queries to Retrieve Information from AdventureWorks

8. How many people work in each department? You will use the following tables and JOIN them:

   HumanResources.EmployeeDepartmentHistory
   HumanResources.Department

   Start working by writing and executing a query that will give you a list of employees and their current department. HINT: This should give you the same number of total rows that you get when selecting all the rows from HumanResources.Employee.

   After you have a good list, modify this to get a count for each department just using the HumanResources.EmployeeDepartmentHistory table. This should give you 16 rows. Then join the HumanResources.Department table to get the department name.

   Include a **grand** total of the number of employees.

   In your result, show the department name (Name) and call it Department, and the number of employees and call this NumberEmployees. This should be in alphabetical order by the department name.

Paste your entire query into Word.
Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #8.

9. Make sure your name is on your document. Submit one document **as a PDF** to Assignment 5.

MSBA 5223 – Database Foundations
Assignment 7

The purpose of this assignment is to help you practice the following skills that are essential to your success in this course and in your professional life beyond school if you have the opportunity to work with databases.

- Code SELECT statements that require any of the language elements presented in this chapter or previous chapters.

Another purpose of this assignment is for you to possibly struggle and feel confused while you apply what you have learned in a slightly different context than what was provided in examples. This will allow you to internalize the use of the various techniques and develop your own approach for addressing problems. The purpose is NOT to make you struggle overmuch and hit your head against a brick wall. If you don't know why something isn't working, ask! If you are not sure why something is doing something other than you expected, ask. I am here to help you learn.

Notes about submitting your assignment:

Instructions on submitting are detailed in assignment 3.

Be sure to save your document periodically. When you have finished the assignment, save the document as a PDF.

Each answer is weighted equally and is worth 12.5 points.

Execute Queries to Retrieve Information from the CentralCollege Database

ALL Columns in every query should have a name.

1. Write a SELECT statement that returns the same result set as this SELECT statement, but don't use a join. Instead, use a subquery in a WHERE clause that uses the IN keyword.

```
SELECT  DISTINCT LastName, FirstName
FROM    Instructors i
        JOIN Courses c
            ON i.InstructorID = c.InstructorID
ORDER BY LastName, FirstName
```

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #1.

2. Write a SELECT statement that answers this question: Which instructors have an annual salary that's greater than the average annual salary for all instructors?

   Use a Subquery.

   Return the LastName, FirstName, and AnnualSalary columns for each Instructor.

   Sort the result set by the AnnualSalary column in descending sequence.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #2.

3. Write a SELECT statement that returns the LastName and FirstName columns from the Instructors table.

   Return one row for each instructor that doesn't have any courses in the Courses table. To do that, **use a subquery introduced with the NOT EXISTS** operator.

   Sort the result set by LastName and then by FirstName.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #3.

4. Write a SELECT statement that returns the LastName and FirstName columns from the Students table, along with a count of the number of courses each student is taking from the StudentCourses table.

   Return one row for each student that is taking more than one class. To do that, **use a subquery with the IN clause** that groups the student course by StudentID.

   You will need a JOIN in both the outer query and in the subquery.

   Group and sort the result set by the LastName and then by the FirstName.

Paste your entire final query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #4.

5. Write a SELECT statement that returns the LastName, FirstName, and AnnualSalary columns of each instructor that has a unique annual salary. In other words, don't include instructors that have the same annual salary as another instructor.
   Use a subquery.
   Sort the results by LastName and then by FirstName.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #5.

6. First, write a SELECT statement that returns one row for each course. It should return these columns:

   The CourseID column from the Courses table

   The most recent enrollment date for that course

   Next, change the SELECT statement to a CTE.

   Then, write a SELECT statement that uses the CTE and returns one row per course that shows the CourseDescription for the course and the LastName, FirstName, and EnrollmentDate for the student with the most recent enrollment data.

Paste your entire FINAL query (including the CTE) ONLY into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #6.

7. First, write a SELECT statement that returns one row for each student who has courses. It should return with these columns:

   The StudentID column from the Students table

   The sum of the course units for that student

   Include only those students who are taking more than 9 units (fulltime).

   Next, change the SELECT statement to a CTE.

   Then, write a SELECT statement that uses this CTE to return the student ID, sum of their course units, and their tuition. (The tuition is equal to the FullTimeCost column, plus the PerUnitCost column multiplied by the number of units.)

   (Hint: You can use a cross join to add the columns from the Tuition table to the query. This works because there's only one row in the Tuition table.)

   Paste your entire FINAL query (including the CTE) ONLY into Word.
   Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #7.

Execute Queries to Retrieve Information from AdventureWorks

8. Use JOINs to list the Department Name (using HumanResources.Department) and the largest pay rate for each **department** (using HumanResources.EmployeePayHistory.)

   Sort the results by this rate descending. Make sure that all columns in the result have a name. Only use employees currently in the department.

   Do NOT use a subquery.

   Paste your entire query into Word.
   Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #8.

9. Submit the one document **as a PDF** to the Assignment - Assignment 7.

MSBA 5223 – Database Foundations
Assignment 8

The purpose of this assignment is to help you practice the following skills that are essential to your success in this course and in your professional life beyond school if you have the opportunity to work with databases.

- Code SELECT statements that require any of the language elements presented in the chapters that have been covered.

Another purpose of this assignment is for you to possibly struggle and feel confused while you apply what you have learned in a slightly different context than what was provided in examples. This will allow you to internalize the use of the various techniques and develop your own approach for addressing problems. The purpose is NOT to make you struggle overmuch and hit your head against a brick wall. If you don't know why something isn't working, ask! If you are not sure why something is doing something other than you expected, ask. I am here to help you learn.

Notes about submitting your assignment:

Instructions on submitting are detailed in assignment 3.

Be sure to save your document periodically. When you have finished the assignment, save the document as a PDF.

Execute Queries to Retrieve Information from the CentralCollege Database

ALL Columns in every query should have a name.

1. Write an INSERT statement that adds a row to the Departments table for a department named "History".

   (Hint: Remember that you do not need to specify a value for identity columns.)

   The Results area of SSMS will say "1 row added." This IS the result you should capture. I do NOT want to see the result of a SELECT query. This is true for all queries that have a result similar to this.

   Paste your entire query into Word.
   Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #1.

2. Write a single INSERT statement that adds two rows to the Instructors table with the following values:

InstructorID	LastName	FirstName	Status	DepartmentChairman	HireDate	AnnualSalary	DepartmentID
Next auto generated ID	Benedict	Susan	P	0	The current Date	34000.00	9
Next auto generated ID	Adams	NULL	F	1	The current Date	66000.00	9

Paste your entire query into Word.
Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #2.

3. Write an UPDATE statement that changes the AnnualSalary column for Susan Benedict (the first instructor you added in exercise 2) from 34,000 to 35,0000. To identify the row, use the InstructorID column.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #3.

4. Write a DELETE statement that deletes Adams (the second instructor you added in exercise 2). To identify the row, use the InstructorID column.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #4.

5. Write a DELETE statement that deletes the row in the Departments table that has an ID of 9.

   Then, run the statement. It will produce an error since the department has related rows in the Instructors table.

   To fix that, precede the DELETE statement with another DELETE statement that deletes all instructors in this department from the Instructors table:

   Execute these two statements **TOGETHER** and paste the ONE Results area to your answer.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #5.

6. Write an UPDATE statement that increases the annual salary for all instructors in the Education department by 5%. (Hint: join the Departments and Instructors tables and then filter the rows by the department **name**.)

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #6.

7. Write a DELETE statement that deletes instructors who aren't teaching any courses.
   To do that, use a subquery in the WHERE clause.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #7.

8. Write an INSERT statement that inserts rows from the Students table into a new table named GradStudents.

   Include only the rows for students who have graduated and do not use a column list.

   Do NOT create the new table before you execute the INSERT statement.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #8.

BEFORE CONTINUING, RESTORE THE CENTRALCOLLEGE DATABASE, just as you did at the beginning of the semester. In the Restore **Options** screen, you will need to UNCHECK the option to take a tail-log backup.

9. Write a SELECT statement that returns these columns from the Instructors table:

MonthlySalary	the AnnualSalary column divided by 12
MonthlySalary1Dec	A column that uses the CAST function to return the monthly salary with 1 digit to the right of the decimal point
MonthlySalarytINT1	A column that uses the CONVERT function to return the monthly salary as an integer
MonthlySalaryINT2	A column that uses the CAST function to return the monthly salary as an integer

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #9.

10. Write a SELECT statement that returns these columns from the Students table:

EnrollmentDate	As is from the Students table
EnrollmentDateOnly	Column that uses the CAST function to return the EnrollmentDate with its date only (year, month, and day) Do not make this more complicated than it needs to be
EnrollmentTimeOnly	Column that uses the CAST function to return the EnrollmentDate with its full time only (hour,minutes, seconds, and milliseconds) Again, not complicated.
EnrollmentYearMonth	Column that uses the CAST function to return the EnrollmentDate with just the year and month. Play with VARCHAR to accomplish this.
EnrollmentMM_DD_YYYY	Column that uses the CONVERT function to return the EnrollmentDate in the format MM/DD/YYYY. In other words, use 2-digit months and days and a 4-digit year, and separate each date component with slashes.
Enrollment12AM_PM	Column that uses the CONVERT function to return the EnrollmentDate with the date, and the hours and minutes on a 12-hour clock with an am/pm indicator
EnrollmentTime24	Column that uses the CONVERT function to return the EnrollmentDate with just the time in a 24-hour format, including milliseconds.
EnrollmentMonthDay	Column that uses the CONVERT function to return the EnrollmentDate with just the month and day such as Mar 05.

Paste your entire query into Word.
Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #10.

11. Make sure your name is on your document. Submit the one document **as a PDF** to the Assignment - Assignment 8.

MSBA 5223 – Database Foundations
Assignment 9

The purpose of this assignment is to help you practice the following skills that are essential to your success in this course and in your professional life beyond school if you have the opportunity to work with databases.

- Code SELECT statements that require any of the language elements presented in this chapter or previous chapters.

Another purpose of this assignment is for you to possibly struggle and feel confused while you apply what you have learned in a slightly different context than what was provided in examples. This will allow you to internalize the use of the various techniques and develop your own approach for addressing problems. The purpose is NOT to make you struggle overmuch and hit your head against a brick wall. If you don't know why something isn't working, ask! If you are not sure why something is doing something other than you expected, ask. I am here to help you learn.

Notes about submitting your assignment:

Instructions on submitting are detailed in assignment 3.

Be sure to save your document periodically. When you have finished the assignment, save the document as a PDF.

Each answer is weighted equally and is worth 12.5 points.

Execute Queries to Retrieve Information from the CentralCollege Database

ALL Columns in every query should have a name.

At the beginning of your Word Document, add a comment of the version of SSMS that you are using. It should look something like --SSMS v20.2.30.0

I am just curious about how some functions may work differently on different versions of SSMS.

DO NOT use conversion functions unless one is specifically mentioned.

1. Write a SELECT statement that returns these columns from the Instructors table:

AnnualSalary	
MonthlySalary	the result of dividing the AnnualSalary column by 12
MonthlySalaryRounded	Calculation of the monthly salary and then use the ROUND function to round the result to 2 decimal places

> Paste your entire query into Word.
> Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #1.

2. Write a SELECT statement that returns these columns from the Students table:

EnrollmentDate	
DateYear	the four-digit year that's stored in the EnrollmentDate column
DateDay	the day of the month that's stored in the EnrollmentDate column
EnrollmentDate4	add four years to the EnrollmentDate column and return only the year. Do NOT use the plus sign to add.

> Paste your entire query into Word.
> Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #2.

3. Write a SELECT statement that returns these columns for each course:

DepartmentName	of the course
CourseNumber	
InstructorName	LastName and FirstName of the instructor concatenated as "LastName, FirstName" such that those without a firstname or lastname are still included.
CourseReference	A column that includes the first three characters from the DepartmentName column in uppercase, concatenated with the CourseNumber column, then either the first character of the FirstName column or an empty string if the column's value is null, and finally the LastName column. (Hint: you will need to cast the CourseNumber column to a character data type.) As an example, this course would look like MSB5223TWilliams.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #3.

4. Write a SELECT statement that returns these columns from the Students table:

StudentName	LastName and FirstName of the student concatenated as "LastName, FirstName" such that those without a firstname or lastname are still included
EnrollmentDate	
GraduationDate	
MonthsToGraduate	The number of months between the EnrollmentDate and GraduationDate columns

Return one row for each student who has graduated.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #4.

5. Write a SELECT statement that answers this question:

What is the total number of courses taught by parttime instructors? Return these columns:

InstructorName	The instructor last name and first name from the Instructors table in this format: "Doe, John". When the instructor's first name is NULL, use FNU as the first name (First Name Unknown). Use the ISNULL() function to accomplish this.
TotalCourses	The total number of courses taught for each instructor.

Use the ROLLUP operator to include a row that gives the grand total. The grand total row should have the InstructorName column (or description) as **Grand Total**. Use the CASE statement to accomplish this.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #5.

6. Write a CTE with a SELECT statement that returns one row for each student who has at least one course. This query should return these columns:

StudentID	
TotalCourseUnits	The sum of the course units for that student

Write a SELECT statement that uses this CTE to return these columns for each student:

StudentID	
TotalCourseUnits	
StudentStatus	Whether the student is fulltime or parttime. To determine whether a student is fulltime, use the IIF function to test if the sum of course units is greater than 9.
TotalTuition	The student's total tuition. To calculate the tuition, use the IIF function to determine whether a student is fulltime or parttime. Then, multiply the sum of course units by the PerUnitCost column in the Tuition table and add that to either the FullTimeCost or PartTimeCost column in the Tuition table. To do that, use a cross join to join the CTE and the Tuition tables. This makes the columns from the Tuition table available to the SELECT statement.

This is NOT the same as the similar query in Assignment 7, but you will get the same answer.

Paste your entire query into Word.
Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #6.

Execute Queries with Functions to Retrieve Information from AdventureWorks

7. Write a SELECT query to show the number of sales for each InvoiceDate along with a running total and a 7-day lag.

   To do this, you will use the Sales.SalesOrderHeader table and write a CTE that will return these columns:

OrderDate	
DayOfWeek	Use the appropriate function to return the numeric day of the week and then use a CASE statement to display it as the name of the day, i.e. Sunday, Monday, etc.
TotalSales	the total number of sales for that day.

   Then, write a SELECT statement, using the CTE, that returns these columns:

OrderDate	
DayWeek	
TotalSales	
RunningTotal	of the TotalSales
7DayLag	based on the OrderDate with a 7-day lag.

   Only return orders for 2014.

   The 7-day lag of the first 7 rows will be NULL. Think about it.

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #7.

8. Write a SELECT query that will return these columns from Sales.vSalesPerson:
   LastName
   TerritoryName
   CountryRegionName

   Do not include rows where the TerritoryName is NULL.

   Sort the results by the TerritoryName when the CountryRegionName is the United States and by the CountryRegionName when it is not. (Hint: Use a CASE statement in the ORDER BY clause.)

Paste your entire query into Word. Use the Snipping Tool to capture the results area of your query. Paste it into your Word document as Answer #8.

9. Make sure your name is on your document. Submit the one document **as a PDF** to the Assignment - Assignment 9.

Using Geography in SQL Server

This assignment will use **AdventureWorks**. We will explore using the capabilities built-in to SQL Server for locations and mapping. This is at an introductory level.

In SSMS, look at the table **Person.Address**. Notice the column **SpatialLocation** with a data type of **geography**. Look at the data in this table. **SpatialLocation** is stored as hexadecimal. It uses ellipsoidal (round-earth) data including GPS latitude and longitude coordinates.

When data is inserted into a geography data type it is encoded using a Spatial Reference Identifier (SRID), a system defined by the European Petroleum Survey Group (EPSG) standard. The data in AdventureWorks was encoded with SRID 4326 which is very common. This standard identifies the units of measure used in the location. SRID uses meters and a specific GPS system.

This assignment will walk you through translating the SpatialLocation in Person.Address into information that can be used by Google maps and then creating a personal Google map containing the home addresses of AdventureWorks employees.

1. The first step is to create a user-defined function. This class does not usually cover this; however, I thought it would add a little something to this assignment. User-defined functions are used the same as built-in functions such as COUNT() and SUM(); however, instead of counting, the function does whatever you have designed it to do. This function will have two parameters, a first name and a last name and will return the names in the format last name, first name, for example **Williams, Therese**. Use the following query code to create the function. Do not be concerned if you have syntax errors. Intellisense is frequently wrong with working with creating functions and procedures. It is true that the CREATE FUNCTION statement must be the first and the only statement in a batch but it is OK to have others before or after it if you only highlight the complete CREATE FUNCTION statement (all of the below) when you execute. There is more information about batches in step 7.

```
CREATE FUNCTION uf_LNFN (@FN nvarchar(50), @LN nvarchar(50)) RETURNS nvarchar(102)
BEGIN
	RETURN RTRIM(@LN) + ', ' + @FN
END
```

2. Test the function in a SELECT query by executing the following code:

```
SELECT dbo.uf_LNFN('John','Smith')
```

DO NOT SUBMIT THE ABOVE AS PART OF THE ASSIGNMENT.

3. (10 points) Use the function to create a list of employees with their name called EmployeeName, the name of their current department called Department, and their job title. Put the columns in the order listed.

Answer #1 **Copy/Paste the query as word keeping the color-coding.** **Use the Snipping Tool to capture the Results area (including the status bar) of your query and paste it.**

4. (10 points) Modify the previous query and **add**, after the previous columns, the City, PostalCode and SpatialLocation of the employee's **home** address. You will need to join three additional tables to do this.

Answer #2 **Copy/Paste the query as word keeping the color-coding.** **Use the Snipping Tool to capture the Results area (including the status bar) of your query and paste it.**

5. (10 points) Modify the previous query and add two columns after the City. The new columns are:

   SpatialLocation.Lat AS Latitude,
   SpatialLocation.Long AS Longitude

   .Lat and **.Long** are built-in methods that will return the Latitude/Longitude of the geography data type.

Answer #3 **Copy/Paste the query as word keeping the color-coding.** **Use the Snipping Tool to capture the Results area (including the status bar) of your query and paste it.**

6. (10 points) Modify the previous query so that instead of displaying the results, they are inserted into a new **temporary** table. Paste the results from executing the query (NOT from a separate SELECT).

Answer #4 **Copy/Paste the query as word keeping the color-coding.** **Use the Snipping Tool to capture the Results area (including the status bar) of your query and paste it.**

7. (10 points) Also, this class does not typically cover making changes to the structure of tables. However, we need to make a change to the temp table. Execute the following query (substituting the name of your temporary table.)

```
ALTER TABLE #t
      ADD Distance float
```

We also want to put information into this column. To do this, we need a variable. For the variable to be active when it is used in the UPDATE query, we will execute a **batch** – multiple statements that must be executed together. Execute the following statements by either highlighting all of them and executing or putting them into a query window by themselves. Make sure you type the following correctly.

```
DECLARE @g geography
SET @g = geography::STGeomFromText('POINT(-122.136626 47.642275)', 4326)
UPDATE #t
      SET Distance = SpatialLocation.STDistance(@g)/1609.344
```

The first statement declares a variable. The second statement uses a provided method to set the value of the variable to a hexadecimal number representing the geography value, using Latitude and Longitude of an address that I decided would be the AdventureWorks Office. It is an address in Redmond, WA. The UPDATE statement changes the value of the Distance column in the temporary table to the difference in distance from the employee's home (SpatialLocation) and this variable. Remember earlier that I said SRID 4326,

which AdventureWorks is using for encoding the geography data type, returns meters for distance. Dividing that value by 1609.344 gives miles.

Select the following columns from your temporary table (in this order);

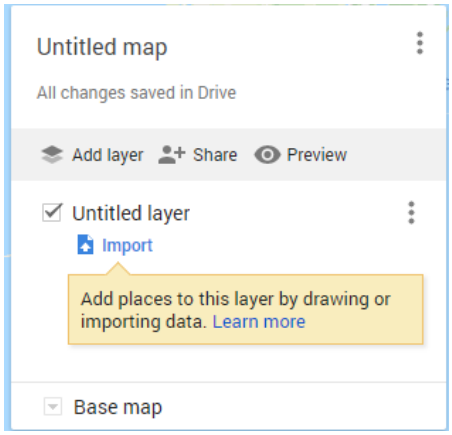
EmployeeName
Department
City
Latitude
Longitude
Distance

Answer #5
Copy/Paste the query as word keeping the color-coding.
Use the Snipping Tool to capture the Results area (including the status bar) of your query and paste it.

8. (10 points) Export the results from the previous query as a .csv file. Open the CSV file with Excel. **Insert** a row at the beginning and add the column names from Answer 5. **Insert** a row after this and add EmployeeName of AdventureWorks, Latitude of 47.642275 and Longitude of -122.136626 and Distance of 0. (This is the office). Save this as a .**CSV**.

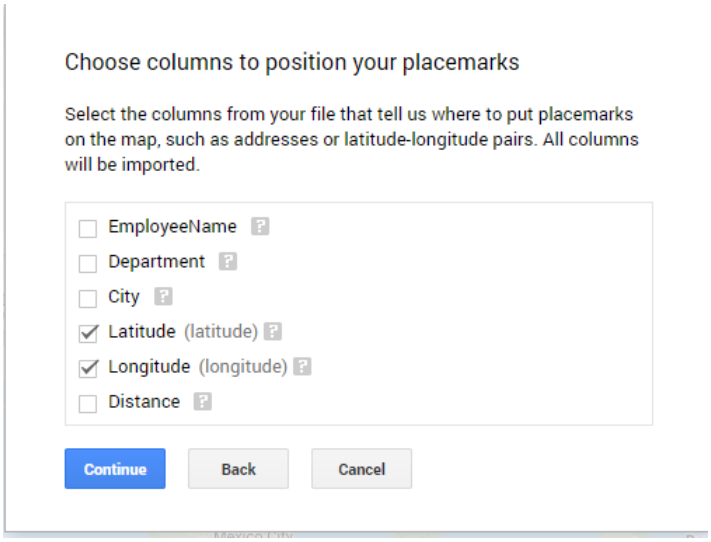
Answer #6
Use the Snipping Tool to capture the first 20 rows or so of the Excel spreadsheet and paste it.

9. (10 points) In a web browser, go to mymaps.google.com. You may have to be logged in with a google account to do this. You will not be able to use your UCO account. Click **Create a New Map**. You should see the following:

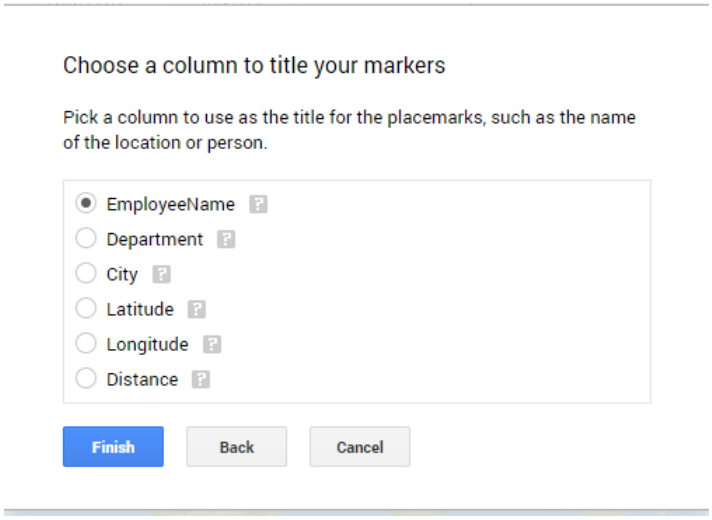


Click Import. Drag your CSV file from Answer 6 to the window.

You should see the following with Latitude and Longitude marked.



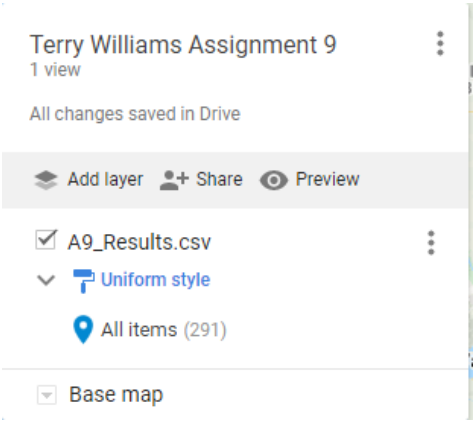
After you have continued, the following will be displayed. Choose EmployeeName to title your markers and Finish.



The map will be loaded with your locations. Click on the marker that is between AR and TN. You will see information about that location.

Answer #7
Use the Snipping Tool to capture ONLY the Popup Window and the icon that goes with it and paste it.

10. (10 points) In the navigation box in the upper left corner, change the name of the Untitled map to your name Assignment **10,** see example below (the example has Assignment 9 but you should have Assignment 10.) In the layer using your CSV file, click ON the style (Uniform style). In the Set Labels dropdown choose EmployeeName as the label. In the Group Places by dropdown, choose Style by Data Column: Department. You will now see a list of departments in the navigation box.



Answer #8
Use the Snipping Tool to capture the entire 'navigation' box and paste it.

11. (10 points) Click on Research and Development and zoom the map in until you can see all the markers for this department. There are 3 in Washington State and 1 in the San Francisco area. Then click on the paint can that will be to the right when you hover over the department in the navigation bar. Choose an icon besides the default. Close the popup box. The last department listed is Other/No Value. This is the AdventuresWorks office. Change the icon to something other than the default. If you go to More Icons, you will find a person on a bicycle under sports! **Zoom in** until you can see this icon along with surrounding area. Also make it a color other than grey so that it is more visible. The map should be where Whidbey Island is still visible in the north and Maury Island is visible in the south, but not much more north or south of that.

Answer #9 **Use the Snipping Tool to capture the entire Google window and paste it.**

12. (10 points) Zoom in closer until you can see the names of employees around the AdventureWorks office.

    Find Lori Kane to the slight southeast and across the lake from AdventureWorks. Click on her icon. Notice the Distance from AdventureWorks (4.43). Underneath the search bar at the top are some icons for tools. Click on the ruler to measure. Click on AdventureWorks, then click twice on the tip of the icon for Lori Kane. Depending on exactly where you clicked, you will see approximately 4.43 miles. Our measure is "as the crow flies." We don't know what algorithms are being used by Google but it is pretty close!

    If you were to get directions for traveling in a car, the distance would be different. Use the icon that looks like a fork in the road next to the ruler to get directions. This will add a layer to your map.

    Click on the Layer Option icon for the directions and choose step-by-step directions. This will give you the distance. Big difference!

    At this point, the map should be showing both the direct measure in a straight line AND the car route following roads.

Answer #10 **Use the Snipping Tool to capture the entire Google window and paste it.**

13. Submit your PDF to Assignment 10.

MSBA 5223 – Database Foundations
Assignment 11

The purpose of this assignment is to help you practice the following skills that are essential to your success in this course and in your professional life beyond school if you have the opportunity to work with databases.

- Code SELECT statements that require any of the language elements in any previous module.

Another purpose of this assignment is for you to possibly struggle and feel confused while you apply what you have learned in a slightly different context than what was provided in examples. This will allow you to internalize the use of the various techniques and develop your own approach for addressing problems. The purpose is NOT to make you struggle overmuch and hit your head against a brick wall. If you don't know why something isn't working, ask! If you are not sure why something is doing something other than you expected, ask. I am here to help you learn.

Notes about submitting your assignment:

Instructions on submitting are detailed in assignment 3.

Be sure to save your document periodically. When you have finished the assignment, save the document as a PDF.

Execute Queries to Retrieve Information from the CentralCollege Database

ALL Columns in every query should have a name.

Execute Queries with Functions to Retrieve Information from AdventureWorks

1. Management at AdventureWorks wants to determine the day of the week which have the most sales in 2012. You will use the Sales.SalesOrderHeader table.

   You will need the following functions – DATENAME(), SUM(), COUNT(), DATEPART().

   List the Day of the Week (based on the OrderDate), Total Revenue (based on the TotalDue), the number of orders and the Revenue per Order.

   Sort the results by the Revenue per Order with the largest first.

   Your results should look like the following:

	DayOfWeek	TotalRevenue	NumberOfOrders	RevenuePerOrder
1	Sunday	8165451.1669	639	12778.4838
2	Tuesday	6739285.0517	618	10904.9919
3	Wednesday	4672792.0455	489	9555.8119
4	Saturday	5087536.6061	545	9334.9295
5	Friday	5126143.8398	577	8884.1314
6	Monday	4870867.6376	589	8269.7243
7	Thursday	3013623.9644	458	6579.9649

Paste your entire query into Word.
Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #1.

2. The Marketing department is preparing an email-blast to notify **individual retail** customers about a clearance sale. You are asked to report on what types of email the customer is willing to receive based on the email addresses within the company's databases.

   The email address counts should be based on email preferences. Email preferences are recorded in the Person.Person table in the column **EmailPromotion**.

   The value 0 indicates "Contact does not wish to receive email promotions."
   The value 1 indicates "Contact does wish to receive email promotions from AdventureWorks."
   The value 2 indicates "Contact does wish to receive email promotions from AdventureWorks and selected partners."

   Use a CASE expression to accomplish this. Do NOT use a UNION.

   Add a grand total row.

   Your results should look like the following:

	EmailPreference	Count
1	Contact does not wish to receive email promotions.	10331
2	Contact does wish to receive email promotions from AdventureWorks.	4667
3	Contact does wish to receive email promotions from AdventureWorks and selected partners.	3486
4	NULL	18484

Paste your entire query into Word.
Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #2.

3. Using SELECT, create a summary table that shows, by territory, the percentage of orders placed online in comparison to retail orders (or orders not placed online).

   The OnlineOrderFlag is 0 for a retail order and 1 for an online order.

   Include the Territory name, the total number of orders, the percentage of orders placed online, the percentage of orders not placed online.

   Display percentages with a % sign without decimals. Look at using FORMAT() to do this. FORMAT is not in your book.

   Use the Sales.SalesOrderHeader and the Sales.SalesTerritory tables.

   You will need to use aggregate functions, and CASE expressions. Some will likely be nested.

   a. Remember when you divide two integers, SQL will give the answer as an integer.
   b. You can use a CASE expression anywhere you use a table column. Think about how aggregates work, i.e. on rows (count the rows, sum a value in a column.)
   c. You do not have to write the entire query at once. Work on small steps. It's easier to see what's going on. Start with a query that simply lists the TerritoryID and TotalOrders. And then work from the outside in (group, add the grandtotal, etc.)

   Your results should look like the following:

	Territory	TotalOrders	RetailOrders	OnlineOrders
1	Australia	6843	1.83%	98.17%
2	Canada	4067	17.01%	82.99%
3	Central	385	97.66%	2.34%
4	France	2672	7.04%	92.96%
5	Germany	2623	5.30%	94.70%
6	Northeast	352	97.16%	2.84%
7	Northwest	4594	11.67%	88.33%
8	Southeast	486	96.50%	3.50%
9	Southwest	6224	12.07%	87.93%
10	United Kingdom	3219	5.84%	94.16%
11	Grand Total	31465	12.10%	87.90%

Paste your entire query into Word.
Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #2.

Use Ranking Functions with AdventureWorks

4. Rank commission percentages by salesperson. A rank of "1" should relate to the salesperson with the greatest commission percentage. If commission percentages are equal among salespeople, rank by Bonus in descending order. Include the Salesperson's name in the format 'lastname, firstname', Commission Percent, Bonus, Rank.

   You will use Sales.SalesPerson and Person.Person.

   Your results should look like the following:

	Salesperson	CommissionPct	Bonus	Rank
1	Pak, Jae	0.02	5150.00	1
2	Mensa-Annan, Tete	0.019	3900.00	2
3	Tsoflias, Lynn	0.018	5650.00	3
4	Valdez, Rachel	0.018	75.00	4
5	Varkey Chudukatil, Ranjit	0.016	985.00	5
6	Saraiva, José	0.015	5000.00	6
7	Carson, Jillian	0.015	2500.00	7
8	Mitchell, Linda	0.015	2000.00	8
9	Blythe, Michael	0.012	4100.00	9
10	Campbell, David	0.012	3500.00	10
11	Reiter, Tsvi	0.01	6700.00	11
12	Ansman-Wolfe, Pamela	0.01	5000.00	12
13	Ito, Shu	0.01	3550.00	13
14	Vargas, Garrett	0.01	500.00	14
15	Jiang, Stephen	0.00	0.00	15
16	Alberts, Amy	0.00	0.00	15
17	Abbas, Syed	0.00	0.00	15

Paste your entire query into Word.
Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #4.

5. Divide employee data into quartiles based on number of months of employment. **By quartile**, provide the following:
   Total number of employees currently in the company;
   Percentage of employees who are Male;
   Percentage of employees who are Female;
   Average number of months of employment as of Jan 1, 2017

   You will need to use either a subquery or a CTE. I have seen this work with the subquery in either the FROM clause or in the JOIN clause, but it should not be both places.

   Also, notice that you need to aggregate rows based on the value of a column: gender.

   You will also use a CASE statement to accomplish this. Think about how aggregates work, i.e. on rows (count the rows, sum a value in a column.) Gender is not quantitative so it cannot be summed. Think about how the CASE statement works, i.e. substituting one value for another.

   Put all that together to aggregate the above information.

   Your results should look like the following:

	Quartile	Employees	%Male	%Female	AvgMonthsEmployed
1	1	73	67.12%	32.88%	76
2	2	73	72.60%	27.40%	94
3	3	72	68.06%	31.94%	96
4	4	72	76.39%	23.61%	100

Paste your entire query into Word.
Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #5.

6. Submit your PDF to Assignment 11.

MSBA 5223 – Database Foundations
Assignment 13

The purpose of this assignment is to help you practice the following skills that are essential to your success in this course and in your professional life beyond school if you have the opportunity to work with databases.

- Code SELECT statements that require any of the language elements in any previous module.

Another purpose of this assignment is for you to possibly struggle and feel confused while you apply what you have learned in a slightly different context than what was provided in examples. This will allow you to internalize the use of the various techniques and develop your own approach for addressing problems. The purpose is NOT to make you struggle overmuch and hit your head against a brick wall. If you don't know why something isn't working, ask! If you are not sure why something is doing something other than you expected, ask. I am here to help you learn.

Notes about submitting your assignment:

Instructions on submitting are detailed in assignment 3.

Be sure to save your document periodically. When you have finished the assignment, save the document as a PDF.

Work on this assignment on your own. You may not work with anyone else, including others in this class, family, personal friends, acquaintances, or co-workers. If you have a question, ask me.

ALL Columns in every query should have a name.

Execute Queries to Retrieve Information from AdventureWorks

1. Today is June 5, 2014. The Sales Manager wants to know how many road bikes were sold last month and for the same month last year.
   You know that the Product Subcategory name is "Road Bikes" and you only want to write one query to get the answer.

   Your output will contain the following columns:

	Year from OrderDate
	Name of the month from OrderDate
TotalQty	Of product from Sales.SalesOrderDetail
TotalRevenue	Calculation using OrderQty, UnitPrice and UnitPriceDiscount

   You will use the tables Sales.SalesOrderHeader, Sales.SalesOrderDetail, Production.Product, and Production.ProductSubCategory

   Sort the results by the quantity for the product in descending order.

   You will have 2 rows in your results.

   Paste your entire query into Word.
   Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #2.

2. After reviewing the numbers provided above, the Sales Manager would like more information on a particular road bike – Product Number BK-R93R-48. How many are currently available for sale?

   List, in this order, the product name and the total QtyOnHand.

   Use the Production,ProductInventory and the Production.Product tables.

   Your results will have one row.

   Paste your entire query into Word.
   Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #2.

3. The Sales Manager is now concerned that there will be enough product of this model on hand for the holiday season at the end of the year. They believe they can sell in excess of 200 during that period.

   You need to produce a report listing the components that are used to manufacture this bicycle along with the QtyOnHand and QtyOnOrder of each component. There will be **14 rows** in the result.

   First write a CTE that will return the total inventory of each product.

   Then write a 2nd CTE that will return the total quantity on order of each product. Do not include orders that arrived before today or after 12-24-2014.

   You should list these columns in this order - Component Name, QtyOnHand, QtyOnOrder

   In the CTEs, you will use either Production.Productinventory OR Purchasing.PurchaseOrderDetail.

   In the main query you will use Production.Product, Production.BillOfMaterials, Production.Product a second time (for the components), and both CTEs as tables.

   Sort the results by the name of the component.

   Your results should be similar to the following:

	productID	Name	QtyOnHand	QtyOnOrder
1	952	Chain	589	720
2	948	Front Brakes	767	7150
3	945	Front Derailleur	853	NULL
4	996	HL Bottom Bracket	970	NULL
5	951	HL Crankset	923	NULL
6	807	HL Headset	782	NULL
7	719	HL Road Frame - Red, 48	NULL	NULL
8	820	HL Road Front Wheel	678	NULL
9	813	HL Road Handlebars	811	NULL
10	940	HL Road Pedal	906	5500
11	828	HL Road Rear Wheel	716	NULL
12	519	HL Road Seat Assembly	821	NULL
13	907	Rear Brakes	723	7150
14	894	Rear Derailleur	847	NULL

   Paste your entire query into Word.
   Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #4.

4. Will there be a problem with the inventory? Why or Why not? If so, how will it be solved? Elaborate and use complete sentences. This answer must be your own thoughts.

Enter your answer into your Word document as Answer #4.

5. How many of the bikes, product number BK-R64Y-40 were sold in Nov/Dec 2013?

   List, in this order, the product name, the Total#Sold, and the TotalRevenue.

   You should have 1 row in your result.

Paste your entire query into Word. **Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #5.**

6. Do you think that the Sales Manager is correct on the sales forecast? Base your answer on evidence. For example compare the growth between 2012 and 2013 or two other comparable time periods in 2013 and 2014. This must be your individual ideas and thoughts. Use complete sentences in your answer and **provide copies of all queries and results used.**

Paste your query(ies) into Word along with your answer. **Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #6.**

7. Which bicycles are the 3 highest sellers based on quantity in the first five months of 2014?

   List, in this order, the product name and the QtySold.

   Your results should have 3 rows.

Paste your entire query into Word. **Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document as Answer #7.**

8. Which bicycles are the 3 highest sellers based on Revenue in the first five months of 2014?

   List, in this order, the product name and the Revenue.

   Your results should have 3 rows.

9. Are the lists in 7 & 8 the same? What is the difference and why? Explain fully. Work on your own. Use complete sentences.

Enter your answer into your Word document as Answer #9.

10. What is the product name, revenue, total cost, profit, and profit margins (in this order) on each model of road bikes in the first five months of 2014?

    The profit margin should be a percentage with 4 decimal places displayed.

    Sort the results with the largest profit margins at the top.

    Do you think these margins are typical? Why or why not?

    Paste your entire query into Word.
    Use the Snipping Tool to capture the Results area of your query. Paste it into your Word document along with your answer to the question as Answer #10.

11. Submit your PDF to Assignment 11.